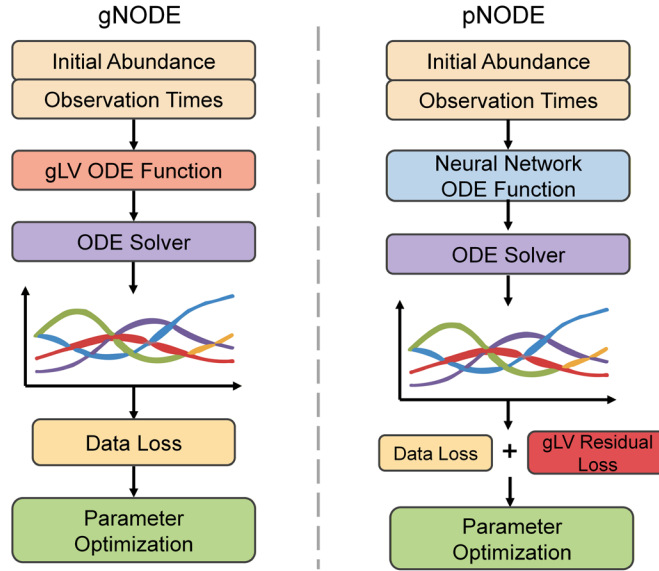
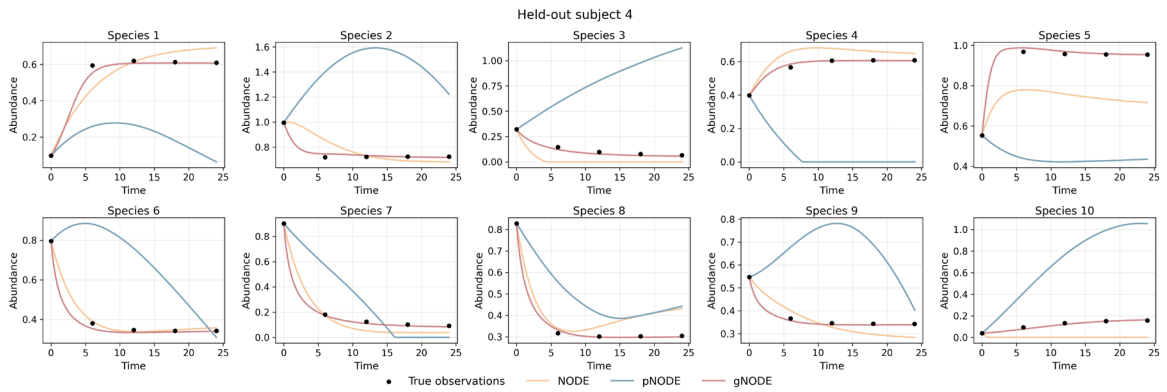


Supplementary Information

Supplementary Figures

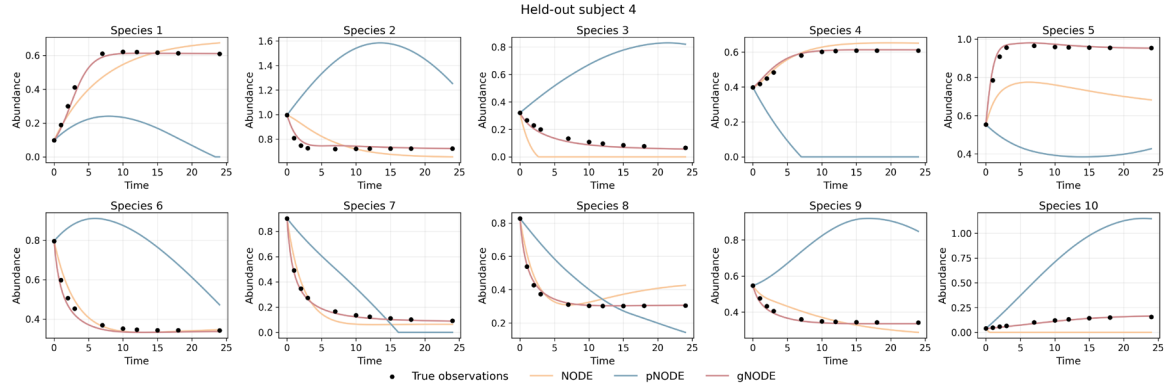


Supplementary Fig. 1 Schematic diagram of the implementation structures of gNODE and pNODE. gNODE directly embeds the gLV equations into the dynamical function of the NeuralODE, whereas pNODE adopts a standard NeuralODE framework and imposes ecological dynamical constraints through an additional gLV residual penalty.

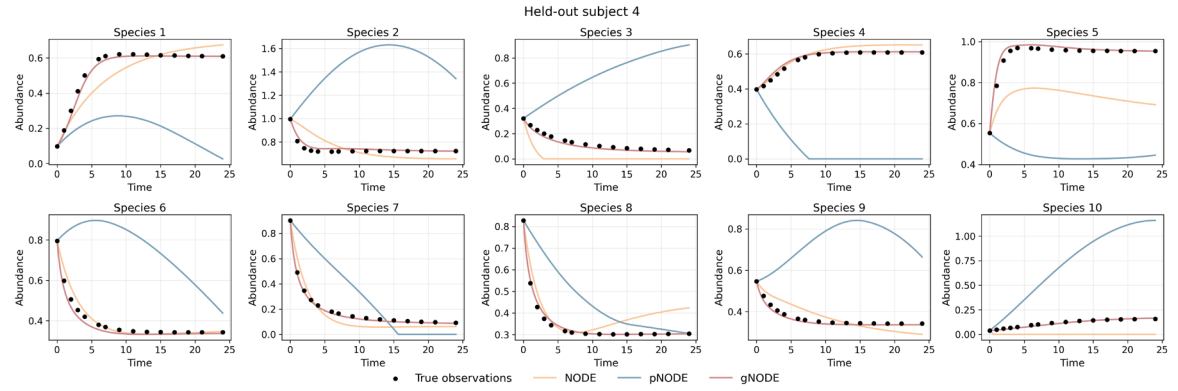


Supplementary Fig. 2 Representative trajectory prediction results of NODE, pNODE, and gNODE on a held-out test sample under the simulation setting with sparsity $\pi = 0.5$, $P = 10$, $S = 10$, data

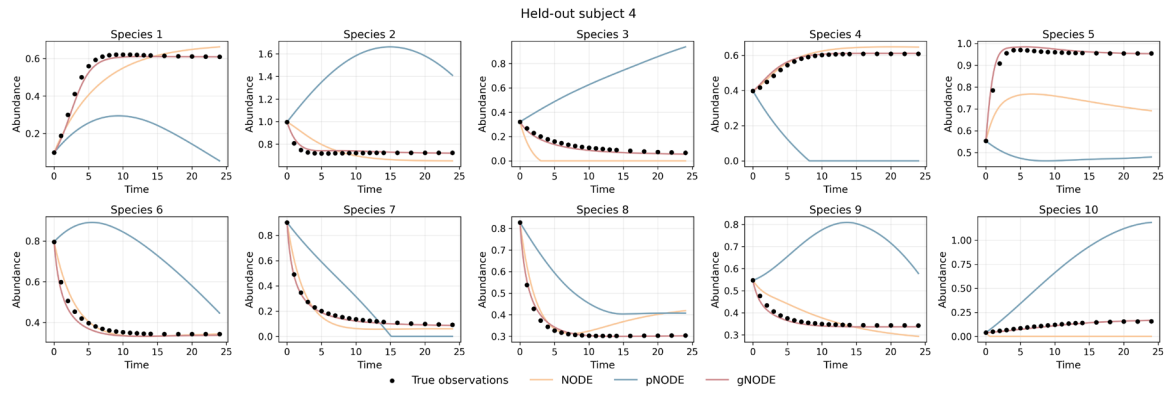
replication = 1, and $T = 5$ time points. The figure shows the first held-out test subject in fold 1 of the five-fold cross-validation, corresponding to subject 4. Each panel represents one species (Species 1–10); black points denote the true observations, and the curves denote the continuous trajectories predicted by NODE, pNODE, and gNODE.



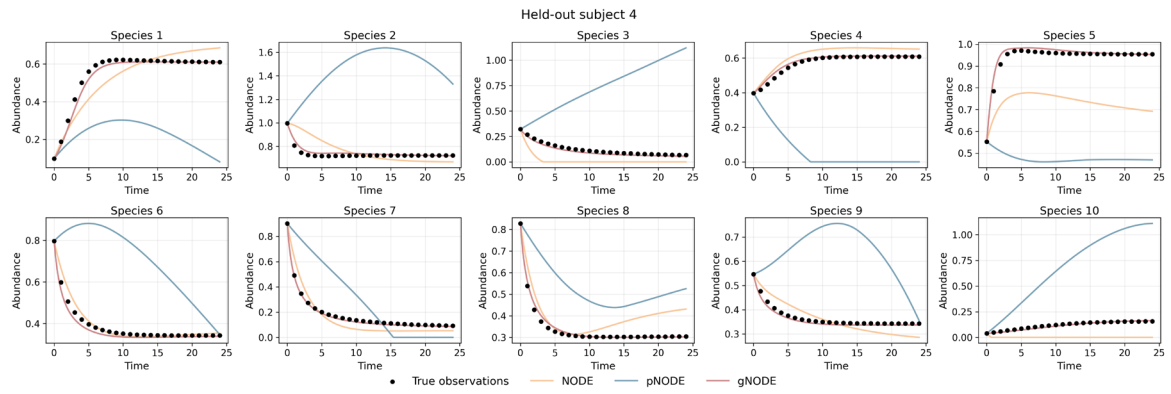
Supplementary Fig. 3 Same as Supplementary Fig. 2, but with $T = 10$.



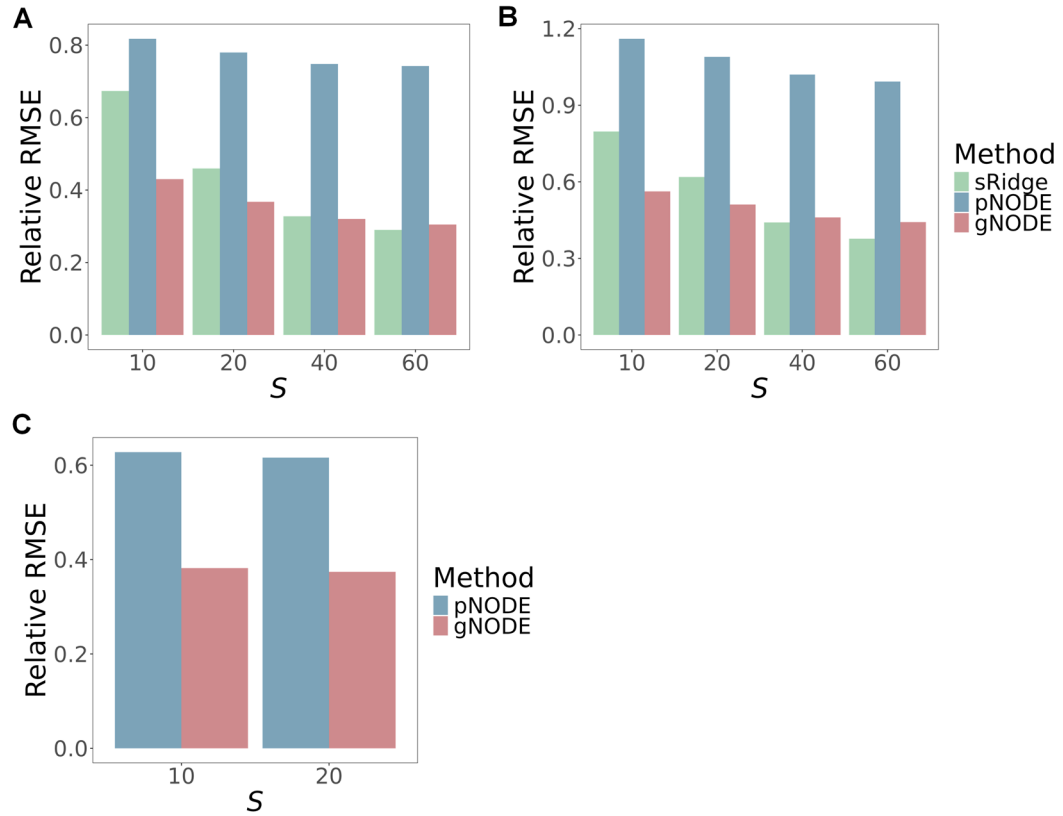
Supplementary Fig. 4 Same as Supplementary Fig. 2, but with $T = 15$.



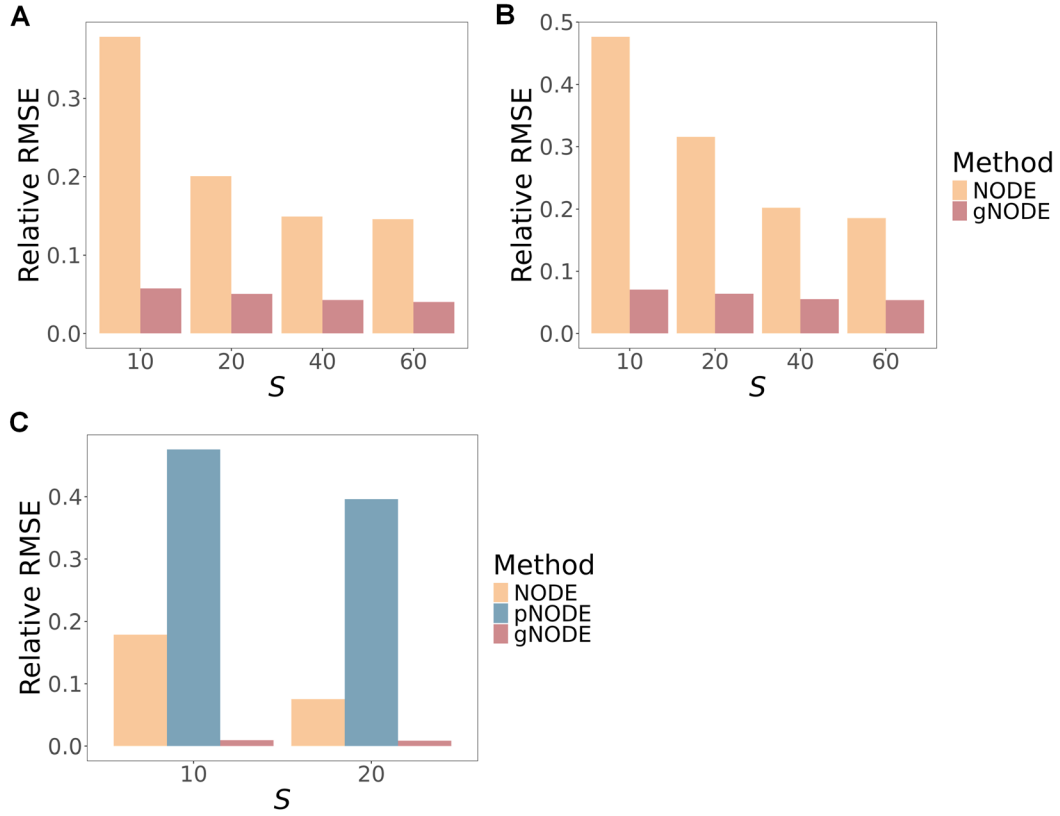
Supplementary Fig. 5 Same as Supplementary Fig. 2, but with $T = 20$.



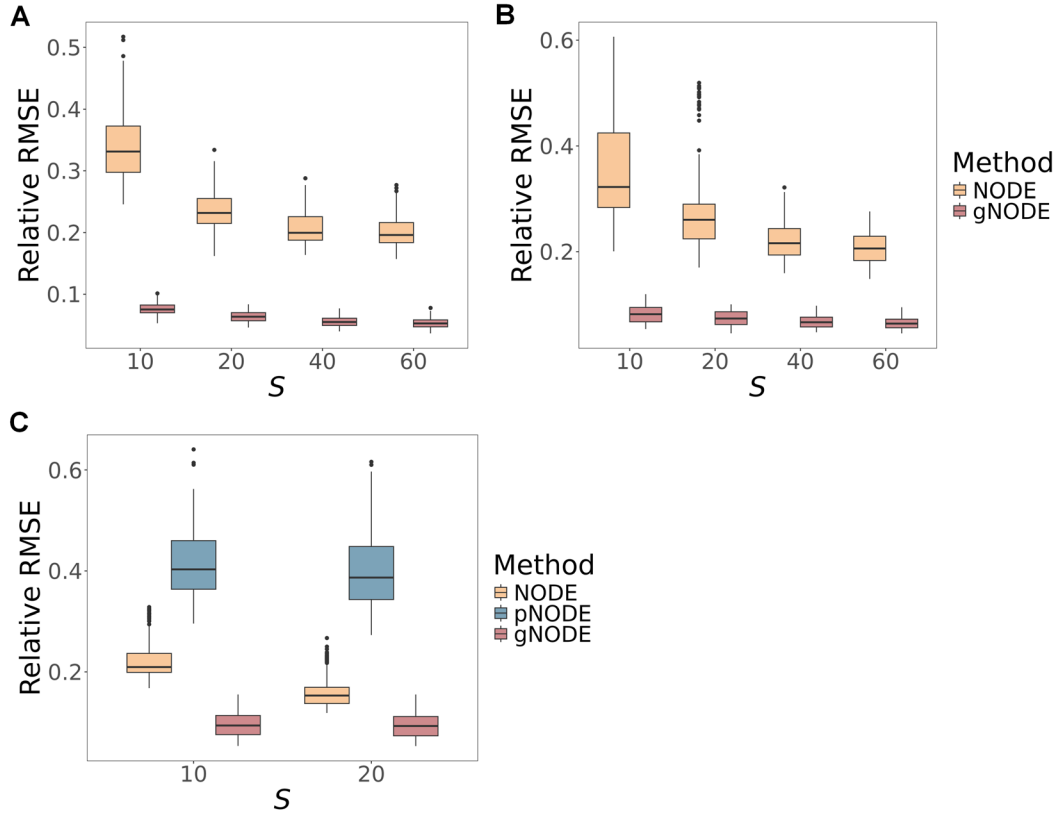
Supplementary Fig. 6 Same as Supplementary Fig. 2, but with $T = 25$.



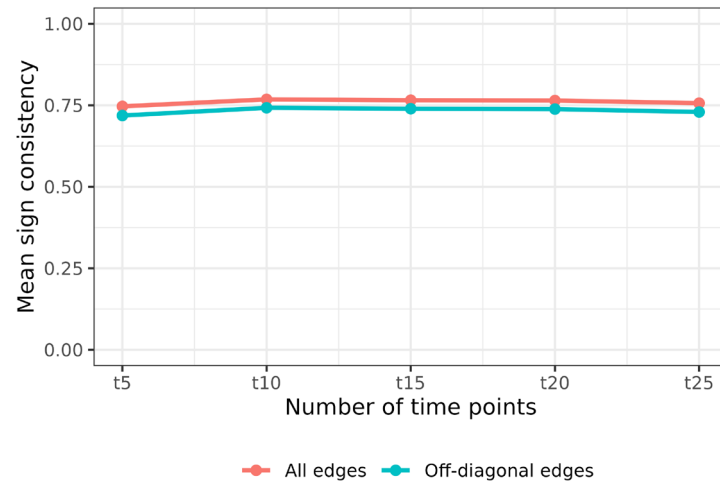
Supplementary Fig. 7 Performance of different methods in parameter estimation. Simulated data generated from the gLV model with a sparsity parameter $\pi = 0.2$ were used to evaluate the estimation performance of different methods under varying numbers of microbes, subjects, and time points. **(A)** Number of microbes $P = 20$, number of subjects $S \in \{10, 20, 40, 60\}$, and number of time points $T = 10$. **(B)** Number of microbes $P = 30$, number of subjects $S \in \{10, 20, 40, 60\}$, and number of time points $T = 10$. **(C)** Number of microbes $P = 10$, number of subjects $S \in \{10, 20\}$, and number of time points $T = 3$. The RMSE was averaged over 10 data replications.



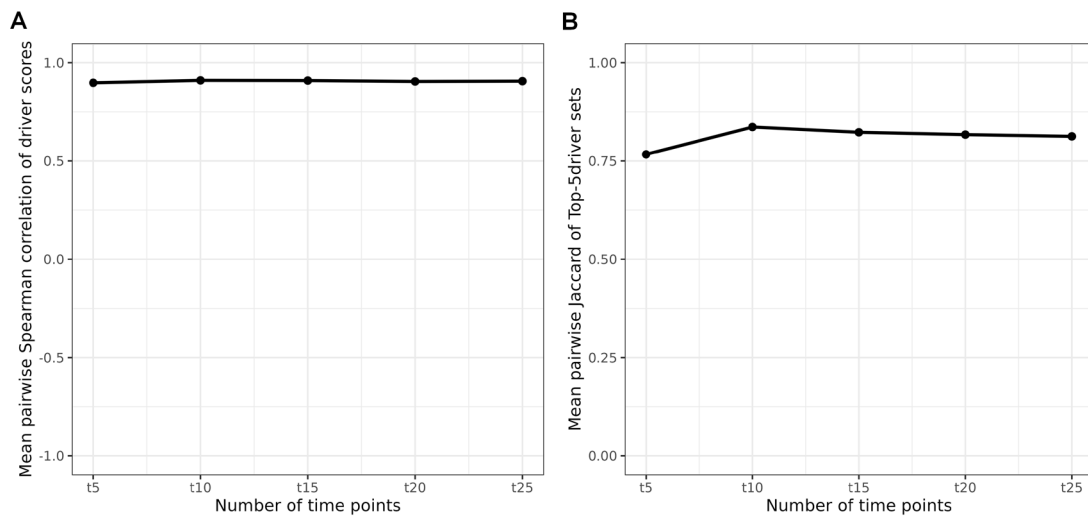
Supplementary Fig. 8 Performance of different methods in trajectory prediction. Simulated data generated from the gLV with a sparsity parameter $\pi = 0.2$ were used to evaluate the prediction performance of different methods under varying numbers of microbes, subjects, and time points. **(A)** Number of microbes $P = 20$, number of subjects $S \in \{10, 20, 40, 60\}$, and number of time points $T = 10$. **(B)** Number of microbes $P = 30$, number of subjects $S \in \{10, 20, 40, 60\}$, and number of time points $T = 10$. **(C)** Number of microbes $P = 10$, number of subjects $S \in \{10, 20\}$, and number of time points $T = 3$. The RMSE was averaged over 10 data replications.



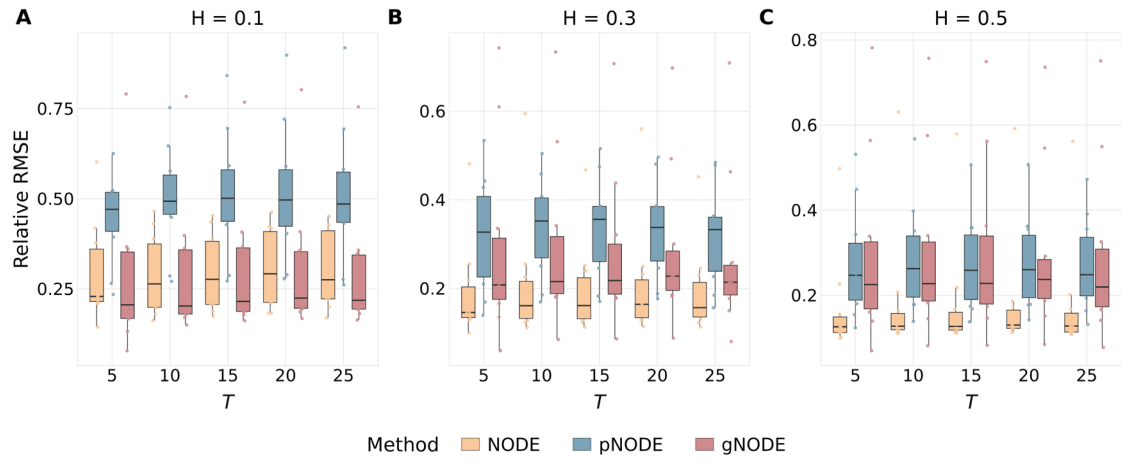
Supplementary Fig. 9 Performance of different methods in predicting intervention trajectories. Simulated data generated from the gLV with a sparsity parameter $\pi = 0.2$ were used to evaluate the prediction performance of different methods under varying numbers of microbes, subjects, and time points. **(A)** Number of microbes $P = 20$, number of subjects $S \in \{10, 20, 40, 60\}$, and number of time points $T = 10$. **(B)** Number of microbes $P = 30$, number of subjects $S \in \{10, 20, 40, 60\}$, and number of time points $T = 10$. **(C)** Number of microbes $P = 10$, number of subjects $S \in \{10, 20\}$, and number of time points $T = 3$. The RMSE was averaged over 10 data replications.



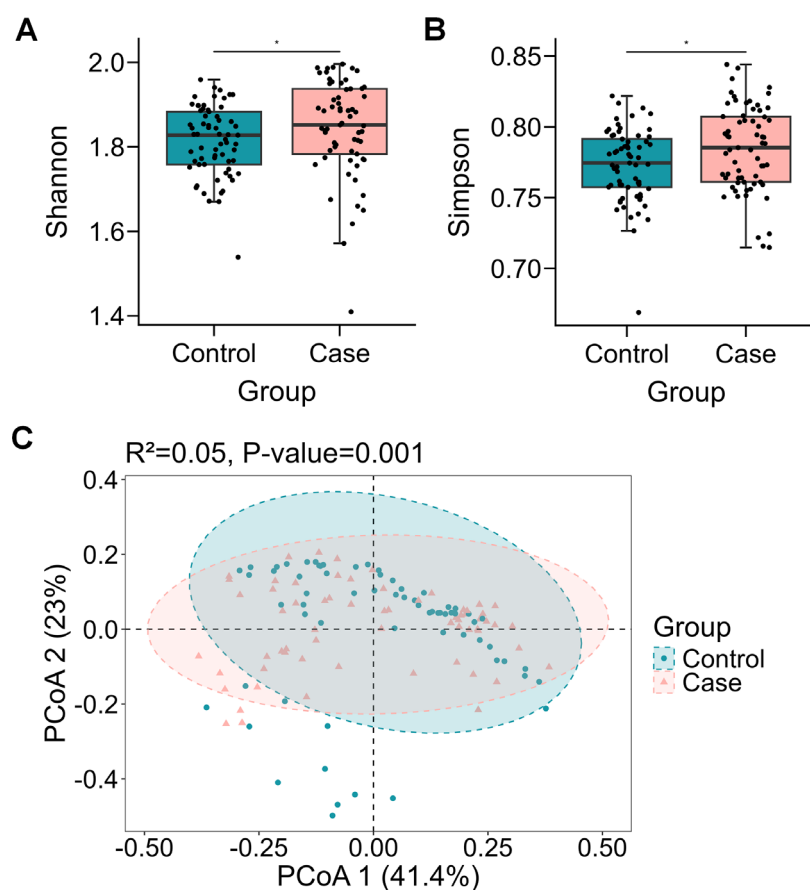
Supplementary Fig. 10 Sign consistency of inferred interaction parameters across different numbers of time points. For each time-point setting, gNODE was refitted 50 times using different random initializations. Mean sign consistency was computed across repeated refits. Results are shown separately for all interaction coefficients and for off-diagonal coefficients only.



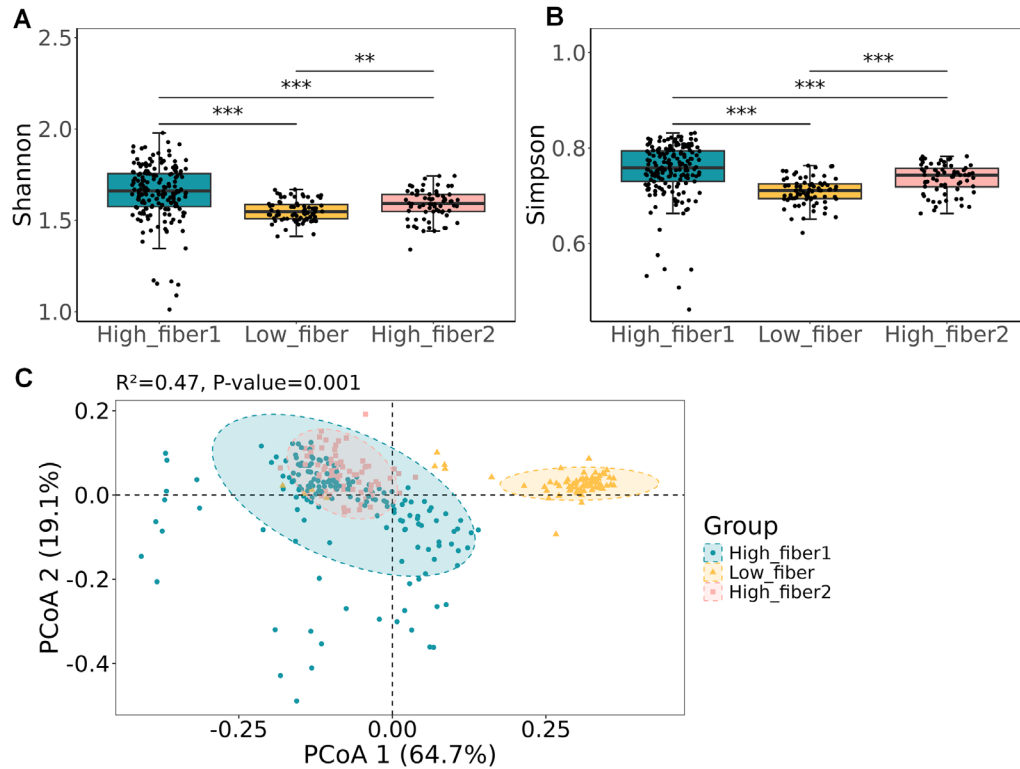
Supplementary Fig. 11 Stability of downstream driver identification across different numbers of time points. For each time-point setting, gNODE was refitted 50 times using different random initializations. **(A)** Mean pairwise Spearman correlation of driver scores across refits. **(B)** Mean pairwise Jaccard index of Top-5 driver sets across refits.



Supplementary Fig. 12 Performance of different methods in trajectory prediction under model misspecification. Simulated data were generated from the nonlinear saturating-response system with a sparsity parameter $\pi = 0.5$, number of microbes $P = 10$, and number of subjects $S = 10$. The prediction performance of NODE, pNODE, and gNODE was evaluated under varying numbers of time points $T \in \{5, 10, 15, 20, 25\}$. **(A)** Saturation constant $H = 0.1$. **(B)** Saturation constant $H = 0.3$. **(C)** Saturation constant $H = 0.5$. The RMSE was averaged over 10 data replications.



Supplementary Fig. 13 Differences in microbial community composition in the *C. difficile* infection dataset. **(A)** α -diversity analysis based on Shannon index. **(B)** α -diversity analysis based on Simpson index. **(C)** β -diversity analysis based on the Bray-Curtis distance. * means P-value < 0.05, ** means P-value < 0.01, and *** means P-value < 0.001.



Supplementary Fig. 14 Differences in microbial community composition in the probiotic cocktail dataset. **(A)** α -diversity analysis based on Shannon index. **(B)** α -diversity analysis based on Simpson index. **(C)** β -diversity analysis based on the Bray-Curtis distance. * means $P\text{-value} < 0.05$, ** means $P\text{-value} < 0.01$, and *** means $P\text{-value} < 0.001$.

Supplementary Tables

Supplementary Table 1. Formal pairwise comparisons of interaction parameter estimation performance among sRidge, pNODE, and gNODE. See Supplementary Table 1 in Excel format.

Supplementary Table 2. Formal pairwise comparisons of trajectory prediction performance among NODE, pNODE, and gNODE. See Supplementary Table 2 in Excel format.

Supplementary Table 3. Formal pairwise comparisons of perturbation response prediction performance among NODE, pNODE, and gNODE. See Supplementary Table 3 in Excel format.

Supplementary Table 4. Empirical variability of inferred off-diagonal interaction coefficients

across repeated gNODE refits.

Time points	Mean empirical 95% interval width of off-diagonal β	Mean SD of off-diagonal β
5	0.0527	0.0927
10	0.0452	0.0795
15	0.0461	0.0811
20	0.0469	0.0826
25	0.0483	0.0849

Supplementary Table 5. Degree metrics of strains under different dietary fiber conditions

Microbe	High fiber	Low fiber	High fiber2
Strain4	11	7	8
Strain6	14	19	20
Strain7	8	7	9
Strain9	14	8	11
Strain13	19	17	16
Strain14	11	9	10
Strain15	13	12	10
Strain16	9	7	11
Strain21	5	10	8
Strain26	13	14	13
Strain27	6	6	6
Strain28	9	9	9
Strain29	10	17	11